Lagrangian Derivation of an HTS Maglev Vehicle with Inerter-Based X-Structure Secondary Suspensions

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Abstract

The four-degree-of-freedom linear suspension equations of Li et al. [1] serve as the reference vehicle model. Each secondary branch in the supplied schematic is replaced by an ideal X-structure whose lateral hinges are joined by a horizontal spring, damper, and inerter in parallel. Exact rigid-rod geometry, static spring preload, the two-terminal inerter energy, and Rayleigh dissipation yield four nonlinear equations, an explicit state-dependent mass matrix, and the small-amplitude limit. Here "exact" applies to the ideal X geometry; the car-body attachment kinematics and primary levitation remain linearized.

1 Connection with the linear vehicle model and sign convention

Li et al. [1], Eqs. (1)–(4), describe car-body heave and pitch and front and rear bogie–Dewar heave. In their notation, m_2, I_2 are the car-body mass and pitch inertia, m_{1a}, m_{1b} are the bogie masses, F_{2i} is the secondary elastic-plus-viscous force, F_{1i} is the dynamic primary force, and f_{ci} is the optional actuator force. For the supplied figure, identify $m_c = m_2$, $I_c = I_2$, $d = L_a = L_b$, $x_c = x_2$, and a, b with the left and right assemblies.

All vertical coordinates $x_c, x_{1a}, x_{1b}, x_a, x_b$ are positive downward. The illustrated φ_c is clockwise positive, so a positive angle raises the left car-body attachment and lowers the right one. The moment signs of Li et al.'s Eq. (2) correspond to the opposite pitch convention: $\theta_2 = -\varphi_c$. Multiplying that reference pitch equation by -1 gives the left and right secondary moment coefficients $-d$ and $+d$, respectively. This mapping fixes every pitch sign below.

The figure is passive, hence $p_a = p_b = 0$. Optional $p_i = f_{ci}$ are retained to show the relation to the reference equations. Define $p_i > 0$ as a downward force on the car body and an equal upward force on bogie i. The reference linear secondary branch, after static-force subtraction, would have $F_{2i}^{\mathrm{lin}} = K_{2i} r_i + C_{2i} \dot{r}_i$.

Assume a rigid car body, small pitch for the attachment-point positions, massless inextensible rods, ideal hinges, and a horizontal spring, damper, and inerter connected between the same two lateral hinges of each X-unit. Bogie pitch, car-body roll and lateral motion, rod mass, joint friction, and nonlinear HTS hysteresis are omitted. The primary levitation is the linearized k_{1i}, c_{1i} model of Li et al.

2 Generalized coordinates and complete IBXS kinematics

Set

$$\boldsymbol{q} = (x_c, \varphi_c, x_{1a}, x_{1b})^{\mathsf{T}}, \quad \boldsymbol{B}_a = (1, -d, -1, 0)^{\mathsf{T}}, \quad \boldsymbol{B}_b = (1, d, 0, -1)^{\mathsf{T}}. \tag{2.1}$$

The perturbation compression of each unit is the downward displacement of its upper terminal relative to its lower terminal:

$$r_a = x_c - d\varphi_c - x_{1a} = \boldsymbol{B}_a^{\mathsf{T}}\boldsymbol{q}, \tag{2.2}$$

$$r_b = x_c + d\varphi_c - x_{1b} = \boldsymbol{B}_b^{\mathsf{T}}\boldsymbol{q}. \tag{2.3}$$

Thus $\dot{r}_i = \boldsymbol{B}_i^{\mathsf{T}}\dot{\boldsymbol{q}}$ and $\ddot{r}_i = \boldsymbol{B}_i^{\mathsf{T}}\ddot{\boldsymbol{q}}$. The r_i are perturbations, not absolute X-unit heights. Inserting r_i directly into an inverse sine would falsely set the equilibrium rod angle to zero.

Let each of the four rods in branch i have length l_i, and let its static angle to the horizontal be $0 < \alpha_i < \pi/2$. The static height, instantaneous height, half-height, half-width, and lateral-hinge span are

$$H_{0i} = 2l_i \sin\alpha_i, \quad H_i = H_{0i} - r_i, \quad s_i = H_i/2, \quad R_i = \sqrt{l_i^2 - s_i^2}, \quad w_i = 2R_i = \sqrt{4l_i^2 - H_i^2}. \tag{2.4}$$

The rigid-rod constraint is $R_i^2 + s_i^2 = l_i^2$. The non-inverted physical branch requires $0 < H_i < 2l_i$, or $H_{0i} - 2l_i < r_i < H_{0i}$. The instantaneous angle obeys $\sin\theta_i = H_i/(2l_i)$ and $\cos\theta_i = w_i/(2l_i)$. Direct differentiation gives the exact Jacobian and acceleration transformation:

$$J_i(r_i) = \frac{\mathrm{d}w_i}{\mathrm{d}r_i} = \frac{H_i}{w_i} = \frac{s_i}{R_i} = \tan\theta_i, \tag{2.5}$$

$$J_i'(r_i) = -\frac{4l_i^2}{w_i^3} = -\frac{l_i^2}{2R_i^3}, \tag{2.6}$$

$$\dot{w}_i = J_i\dot{r}_i, \qquad \ddot{w}_i = J_i\ddot{r}_i + J_i'\dot{r}_i^2. \tag{2.7}$$

Compression reduces height and increases span, hence $J_i > 0$. If an extension coordinate $h_i = -r_i$ is used instead, the same geometry gives $\dot{w}_i = -\tan\theta_i\dot{h}_i$. These sign conventions must not be mixed.

3 Static load and spring preload

For a horizontal spring of stiffness k_{Xi} and natural length w_{ni},

$$V_{Xi}(r_i) = \frac{1}{2}k_{Xi}(w_i - w_{ni})^2, \qquad S_i(r_i) = V_{Xi}'(r_i) = k_{Xi}(w_i - w_{ni})J_i. \tag{3.1}$$

Let $w_{0i} = 2l_i\cos\alpha_i$ and $P_i = S_i(0)$. With no static actuator bias or applied pitch moment, car-body force and moment balance imply

$$P_a + P_b = m_c g, \qquad -dP_a + dP_b = 0, \qquad P_a = P_b = \frac{m_c g}{2}. \tag{3.2}$$

The spring therefore needs the natural length

$$w_{ni} = w_{0i} - \frac{P_i}{k_{Xi}\tan\alpha_i}. \tag{3.3}$$

If $w_{ni} = w_{0i}$ and there is no separate static-load path, the spring supplies zero vertical force at $r_i = 0$ and cannot support the car body. Natural length, allowable stroke, and strength must be feasible. If L_{0i} denotes the static upward primary levitation force on bogie i, its static balance also requires

$$L_{0i} = m_{1i}g + P_i. \tag{3.4}$$

4 Kinetic and potential energy, dissipation, and Lagrange's equation

The horizontal inerter is a two-terminal element, with kinetic energy and vertical equivalent inertance

$$T_{bi} = \frac{1}{2} b_i \dot{w}_i^2 = \frac{1}{2} \mu_i(r_i) \dot{r}_i^2, \quad \mu_i = b_i J_i^2 = b_i \frac{H_i^2}{4l_i^2 - H_i^2}. \tag{4.1}$$

Its derivative is

$$\mu_i' = 2 b_i J_i J_i' = -\frac{8 b_i l_i^2 H_i}{w_i^4} = -\frac{b_i l_i^2 s_i}{R_i^4}. \tag{4.2}$$

The horizontal damper's Rayleigh function is

$$\mathcal{D}_{Xi} = \frac{1}{2} c_{Xi} \dot{w}_i^2 = \frac{1}{2} c_{Xi} J_i^2 \dot{r}_i^2. \tag{4.3}$$

The equivalent vertical damping coefficient is $c_{Xi} J_i^2$, not c_{Xi}.

Let $x_a(t), x_b(t)$ be the guideway inputs. The linearized primary spring energy and dissipation are $k_{1i}(x_{1i} - x_i)^2/2$ and $c_{1i}(\dot{x}_{1i} - \dot{x}_i)^2/2$. After using Eqs. (3.2)–(3.4) to cancel static gravity and levitation terms, the vehicle energies and Rayleigh function are

$$T = \tfrac{1}{2} m_c \dot{x}_c^2 + \tfrac{1}{2} I_c \dot{\varphi}_c^2 + \sum_{i=a,b} \left[\tfrac{1}{2} m_{1i} \dot{x}_{1i}^2 + \tfrac{1}{2} \mu_i(r_i) \dot{r}_i^2 \right], \tag{4.4}$$

$$\widetilde{V} = \sum_{i=a,b} \left[V_{Xi}(r_i) - V_{Xi}(0) - P_i r_i + \tfrac{1}{2} k_{1i} (x_{1i} - x_i)^2 \right], \tag{4.5}$$

$$\mathcal{D} = \sum_{i=a,b} \left[\tfrac{1}{2} c_{Xi} J_i^2 \dot{r}_i^2 + \tfrac{1}{2} c_{1i} (\dot{x}_{1i} - \dot{x}_i)^2 \right]. \tag{4.6}$$

The $-P_i r_i$ term records the canceled static load; it is not an installed negative-stiffness element.

With $\mathcal{L} = T - \widetilde{V}$, Lagrange's equation with Rayleigh dissipation reads

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \frac{\partial \mathcal{L}}{\partial q_j} + \frac{\partial \mathcal{D}}{\partial \dot{q}_j} = Q_j, \qquad \boldsymbol{Q} = p_a \boldsymbol{B}_a + p_b \boldsymbol{B}_b. \tag{4.7}$$

For each inerter, the key terms are

$$\frac{\partial T_{bi}}{\partial \dot{\boldsymbol{q}}} = \mu_i \dot{r}_i \boldsymbol{B}_i, \tag{4.8}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial T_{bi}}{\partial \dot{\boldsymbol{q}}} = \boldsymbol{B}_i (\mu_i \ddot{r}_i + \mu_i' \dot{r}_i^2), \tag{4.9}$$

$$\frac{\partial T_{bi}}{\partial \boldsymbol{q}} = \tfrac{1}{2} \boldsymbol{B}_i \mu_i' \dot{r}_i^2. \tag{4.10}$$

Their difference is $\boldsymbol{B}_i(\mu_i \ddot{r}_i + \mu_i' \dot{r}_i^2/2)$, including the geometric velocity-squared term. The remaining derivatives are

$$\frac{\partial \widetilde{V}_{Xi}}{\partial \boldsymbol{q}} = \boldsymbol{B}_i (S_i - P_i), \qquad \frac{\partial \mathcal{D}_{Xi}}{\partial \dot{\boldsymbol{q}}} = \boldsymbol{B}_i c_{Xi} J_i^2 \dot{r}_i. \tag{4.11}$$

Define the dynamic secondary two-terminal force and primary restoring force as

$$F_i = \mu_i \ddot{r}_i + \tfrac{1}{2} \mu_i' \dot{r}_i^2 + c_{Xi} J_i^2 \dot{r}_i + S_i(r_i) - P_i, \tag{4.12}$$

$$G_i = k_{1i}(x_{1i} - x_i) + c_{1i}(\dot{x}_{1i} - \dot{x}_i). \tag{4.13}$$

Using exact geometry, the secondary force is also

$$\boxed{\begin{aligned} F_i = b_i \frac{H_i^2}{w_i^2}\ddot{r}_i - \frac{4b_i l_i^2 H_i}{w_i^4}\dot{r}_i^2 + c_{Xi}\frac{H_i^2}{w_i^2}\dot{r}_i \\ + k_{Xi}(w_i - w_{ni})\frac{H_i}{w_i} - P_i. \end{aligned}} \tag{4.14}$$

The negative μ_i' and velocity-squared coefficient follow from defining r_i as compression. An extension coordinate reverses the coordinate sign without changing the physics.

5 Four complete equations of motion

Substitution of Eqs. (2.1) and (4.12)–(4.13) into Eq. (4.7) gives

$$\boxed{m_c\ddot{x}_c + F_a + F_b = p_a + p_b,} \tag{5.1}$$

$$\boxed{I_c\ddot{\varphi}_c - dF_a + dF_b = -dp_a + dp_b,} \tag{5.2}$$

$$\boxed{m_{1a}\ddot{x}_{1a} + G_a - F_a = -p_a,} \tag{5.3}$$

$$\boxed{m_{1b}\ddot{x}_{1b} + G_b - F_b = -p_b.} \tag{5.4}$$

For the passive schematic, set $p_a = p_b = 0$. Each F_i contains $\ddot{r}_i$, so these equations are coupled implicitly in acceleration. With $b_i = 0$ and $F_i = K_{2i}r_i + C_{2i}\dot{r}_i$, their force topology reduces to Li et al.'s Eqs. (1), (3), and (4). Their pitch equation (2) is recovered after $\theta_2 = -\varphi_c$.

6 Explicit state-dependent mass matrix

Write

$$N_i = \tfrac{1}{2}\mu_i'\dot{r}_i^2 + c_{Xi}J_i^2\dot{r}_i + S_i(r_i) - P_i, \qquad F_i = \mu_i\ddot{r}_i + N_i. \tag{6.1}$$

With $\boldsymbol{e}_a = (0,0,1,0)^{\mathsf{T}}$ and $\boldsymbol{e}_b = (0,0,0,1)^{\mathsf{T}}$, the acceleration equation suitable for integration is

$$\boxed{\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} = \boldsymbol{B}_a(p_a - N_a) + \boldsymbol{B}_b(p_b - N_b) - \boldsymbol{e}_a G_a - \boldsymbol{e}_b G_b,} \tag{6.2}$$

where

$$\boldsymbol{M} = \operatorname{diag}(m_c, I_c, m_{1a}, m_{1b}) + \mu_a\boldsymbol{B}_a\boldsymbol{B}_a^{\mathsf{T}} + \mu_b\boldsymbol{B}_b\boldsymbol{B}_b^{\mathsf{T}}. \tag{6.3}$$

In the selected coordinate order,

$$\boldsymbol{M} = \begin{bmatrix} m_c + \mu_a + \mu_b & d(\mu_b - \mu_a) & -\mu_a & -\mu_b \\ d(\mu_b - \mu_a) & I_c + d^2(\mu_a + \mu_b) & d\mu_a & -d\mu_b \\ -\mu_a & d\mu_a & m_{1a} + \mu_a & 0 \\ -\mu_b & -d\mu_b & 0 & m_{1b} + \mu_b \end{bmatrix}. \tag{6.4}$$

For positive physical masses and $b_i \geq 0$, $\boldsymbol{z}^{\mathsf{T}}\boldsymbol{M}\boldsymbol{z} = \boldsymbol{z}^{\mathsf{T}}\boldsymbol{M}_0\boldsymbol{z} + \sum_i \mu_i(\boldsymbol{B}_i^{\mathsf{T}}\boldsymbol{z})^2 > 0$ for any nonzero $\boldsymbol{z}$ inside the physical stroke. Thus $\boldsymbol{M}$ is symmetric positive definite. A numerical step evaluates $r_i, H_i, w_i, J_i, \mu_i, N_i, G_i$ from the state and solves Eq. (6.2) for $\ddot{\boldsymbol{q}}$.

7 Small-amplitude limit, stability, and energy check

At $r_i = \dot{r}_i = 0$, $J_{i0} = \tan\alpha_i$. The local vertical inertance and damping are

$$\mu_{i0} = b_i\tan^2\alpha_i, \qquad C_{Xi}^{\tan} = c_{Xi}\tan^2\alpha_i. \tag{7.1}$$

Since $S_i' = k_{Xi}[J_i^2 + (w_i - w_{ni})J_i']$, the tangent stiffness is

$$K_{Xi}^{\tan} = k_{Xi}\left[\tan^2\alpha_i - \frac{w_{0i} - w_{ni}}{2l_i\cos^3\alpha_i}\right]$$
$$= k_{Xi}\tan^2\alpha_i - \frac{P_i}{2l_i\sin\alpha_i\cos^2\alpha_i}. \tag{7.2}$$

Thus $F_i \simeq \mu_{i0}\ddot{r}_i + C_{Xi}^{\tan}\dot{r}_i + K_{Xi}^{\tan}r_i$. Positive local branch stiffness requires $P_i < 2k_{Xi}l_i\sin^3\alpha_i$; full stroke, joint forces, and end stops need separate checks. With $b_i = 0$ and $K_{2i} = K_{Xi}^{\tan}$, $C_{2i} = C_{Xi}^{\tan}$, the first-order vehicle equations coincide with the linear spring-damper model. For $b_i > 0$, the added mass matrix $\sum_i \mu_{i0}\boldsymbol{B}_i\boldsymbol{B}_i^{\mathsf{T}}$ remains, so stiffness and damping matching alone is insufficient.

For fixed guideway inputs and $p_i = 0$, summing the Lagrange equations multiplied by $\dot{q}_j$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(T + \widetilde{V}) = -\sum_{i=a,b}\left(c_{Xi}J_i^2\dot{r}_i^2 + c_{1i}\dot{x}_{1i}^2\right) \le 0. \tag{7.3}$$

This independently checks the sign and J_i^2 factor of the transformed horizontal damper.

8 Model limits

The derivation excludes roll, lateral motion, bogie pitch, rod mass, joint friction, impacts, and hysteretic HTS levitation. The primary suspension is an operating-point linearization inherited from Li et al. If a physical IBXS damper is vertical rather than connected between the lateral hinges, replace Eq. (4.3) by $c_{Xi}\dot{r}_i^2/2$; Eq. (4.12) then contains $c_{Xi}\dot{r}_i$, while the spring and inerter geometry is unchanged. Different terminal locations for the three horizontal elements require separate kinematic Jacobians.

References

[1] Q. Li, Z. Deng, L. Wang, H. Li, J. Zhang, and E. F. Rodriguez, "Active vibration control of secondary suspension based on high-temperature superconducting maglev vehicle system," *Physica C*, vol. 585, article 1353872, 2021. doi:10.1016/j.physc.2021.1353872.